

Pulmonary circulation

Pulmonary circulation

◆ The lung has two circulations:-

i. **High-pressure, low-flow circulation**

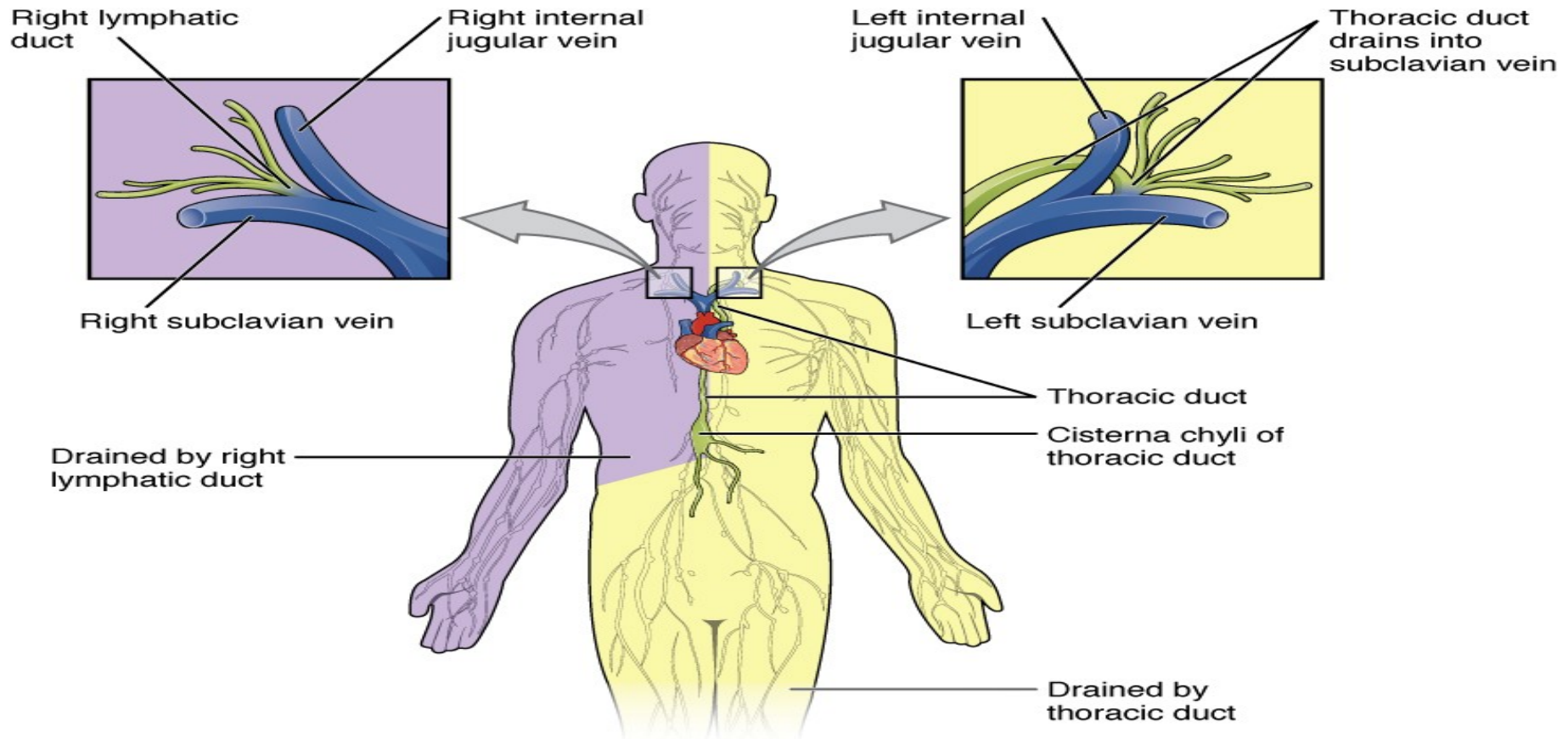
- ✓ supplies **systemic arterial blood** to trachea, *bronchial tree*, supporting tissues of the lung, and outer coats (adventitia) of pulmonary arteries and veins.
- ✓ supply most of this systemic arterial blood at a pressure that is only slightly lower than the **aortic pressure**.

ii. **Low-pressure, high-flow circulation**

- ✓ **supplies venous blood** from all parts of the body to the alveolar capillaries
- ❖ **Bronchial Vessels.** Blood also flows to the lungs through small bronchial arteries that originate from the systemic circulation, amounting to 1 to 2 percent of the total cardiac output.
 - This bronchial arterial blood is *oxygenated* blood, in contrast to the partially deoxygenated blood in the pulmonary arteries.
 - It supplies the supporting tissues of the lungs, including the connective tissue, septa, and large and small bronchi.

Lymphatic

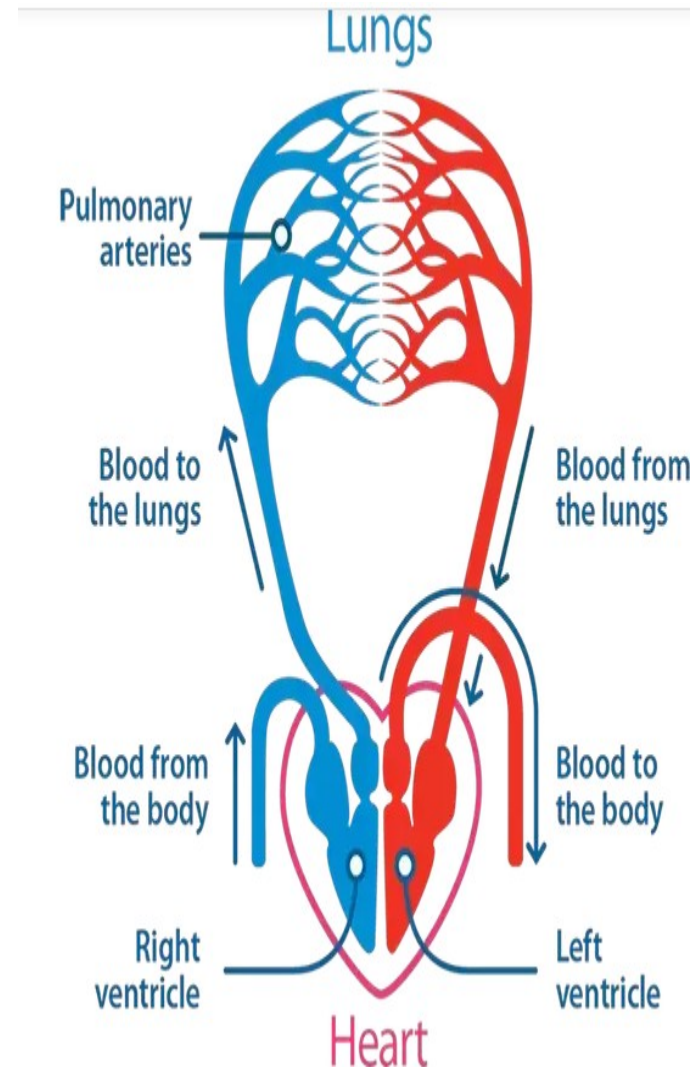
Lymph vessels are present in all the supportive tissues of the lung →
right thoracic lymph duct
Drain particulate matter entering alveoli and leaking plasma proteins
(Prevent pulmonary edema)



Pulmonary circulation

Blood flow through the lungs and its distribution

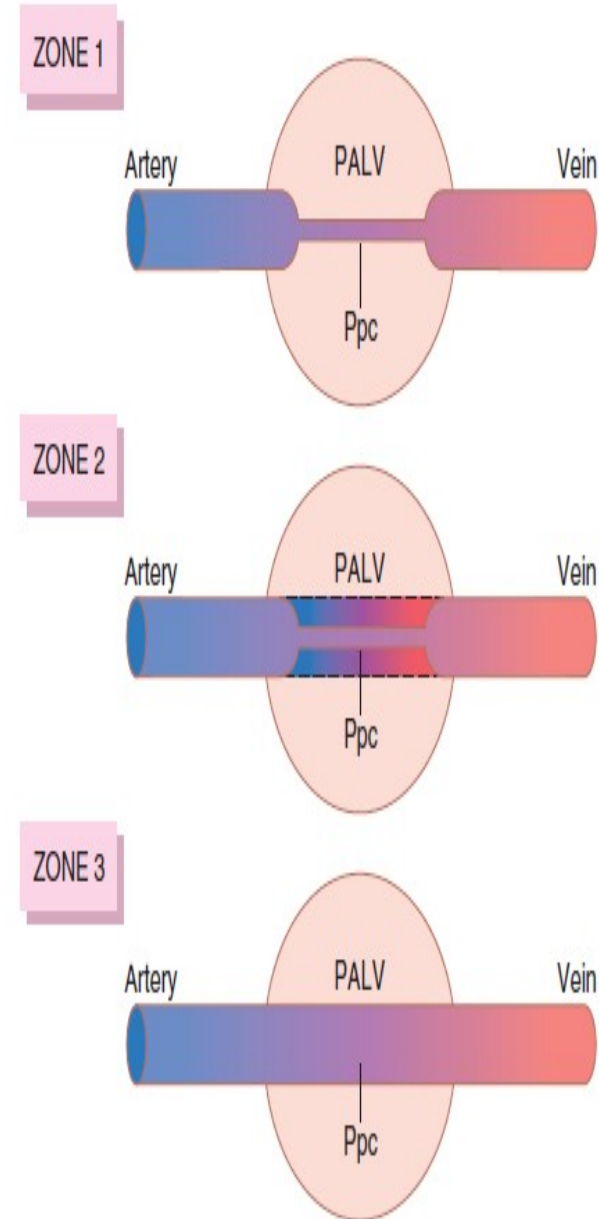
- ◆ The blood flow through the lungs is **essentially equal to the cardiac output**
- ◆ The factors that control CO **also control pulmonary blood flow**
- ◆ Decreased alveolar oxygen **reduces local alveolar** blood flow and regulates pulmonary blood flow distribution (*opposite to the effect observed in systemic vessels*)



Pulmonary circulation

Effect of **Hydrostatic Pressure** gradients in the lungs on regional pulmonary blood flow

- ◆ Under different normal and pathological lung conditions, one may find any one of three possible zones (patterns) of pulmonary blood flow:
 - i. **Zone 1: No blood flow**
 - ✓ alveolar air pressure is greater than arterial pressure
 - ii. **Zone 2: Intermittent blood flow**
 - ✓ systolic arterial pressure rises higher than alveolar air pressure, but diastolic arterial pressure falls below alveolar air pressure
 - iii. **zone 3: continuous flow**
 - ✓ arterial pressure and pulmonary capillary pressure (P_{pc}) remain greater than alveolar air pressure at all times.



Pulmonary circulation

- ◆ Normally, the lungs have only zones 2 and 3 blood flow
 - ✓ Zone:2 (**intermittent flow**) in the apices and
 - ✓ Zone:3 (**continuous flow**) in all the lower areas.
- ◆ **Exercise : increases blood flow** through all parts of the lungs (**pulmonary vascular pressures rise** enough during exercise to convert the *lung apices from a zone 2 pattern into a zone 3*).

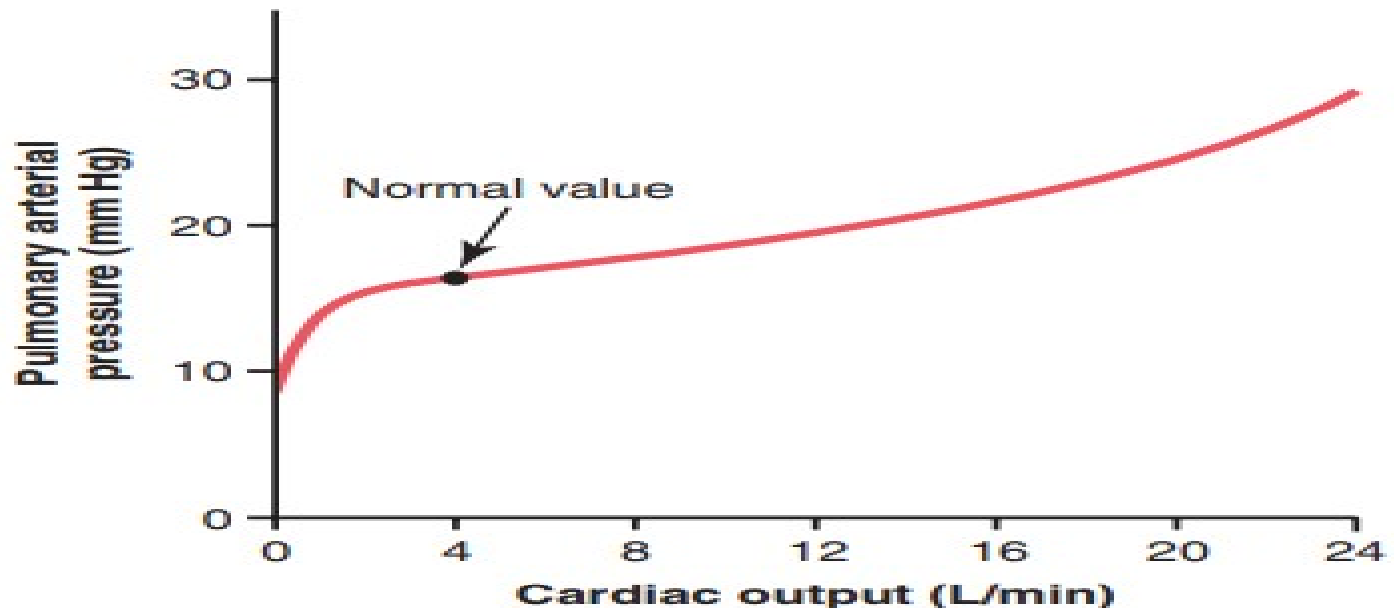


Figure 39-5. Effect on mean pulmonary arterial pressure caused by increasing the cardiac output during exercise.

Cont.....

- **This extra flow is accommodated in the lungs in three ways:**

- (1) By increasing the number of open capillaries, sometimes as much as threefold;
 - (2) By distending all the capillaries and increasing the rate of flow through each capillary more than twofold; and
 - (3) By increasing the pulmonary arterial pressure.
- ❖ Normally, the first two changes decrease pulmonary vascular resistance so much that the pulmonary arterial pressure rises very little, even during maximum exercise.
 - ❖ This effect the ability of the lungs to accommodate greatly increased blood flow during exercise without increasing the pulmonary arterial pressure conserves the energy of the right side of the heart.
 - ❖ This ability also prevents a significant rise in pulmonary capillary pressure and the development of pulmonary edema.